AWS IAM & EC2 Security Access Control Project Report

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GitHub Repository: [https://github.com/atharshareef7/aws-iam-ec2-security-project]

1. Executive Summary

This project demonstrates the implementation of AWS Identity and Access Management (IAM) and Elastic Compute Cloud (EC2) to enforce role-based access control in a simulated DevOps environment. Production and development EC2 instances were created, IAM policies were defined to restrict user access based on environment tags, and permission boundaries were tested to ensure compliance and environment isolation.

2. Objectives

- Launch EC2 instances for production and development environments.
- Create IAM policies restricting access based on resource tags.
- Set up IAM users and groups with appropriate permissions.
- Test permissions for both production and development instances.
- Demonstrate security best practices such as the Principle of Least Privilege.

3. Tools and Services Used

Tool / Service	Purpose
AWS EC2	Launch and manage virtual machines.
AWS IAM	Manage users, groups, and permissions.
AWS Management Console	Graphical interface for AWS operations.
IAM Policy JSON	Defines access control rules.
AWS Account Alias	Simplifies IAM user login process.

4. Implementation Steps

Step 1: Created an Account Alias for easier IAM user login.

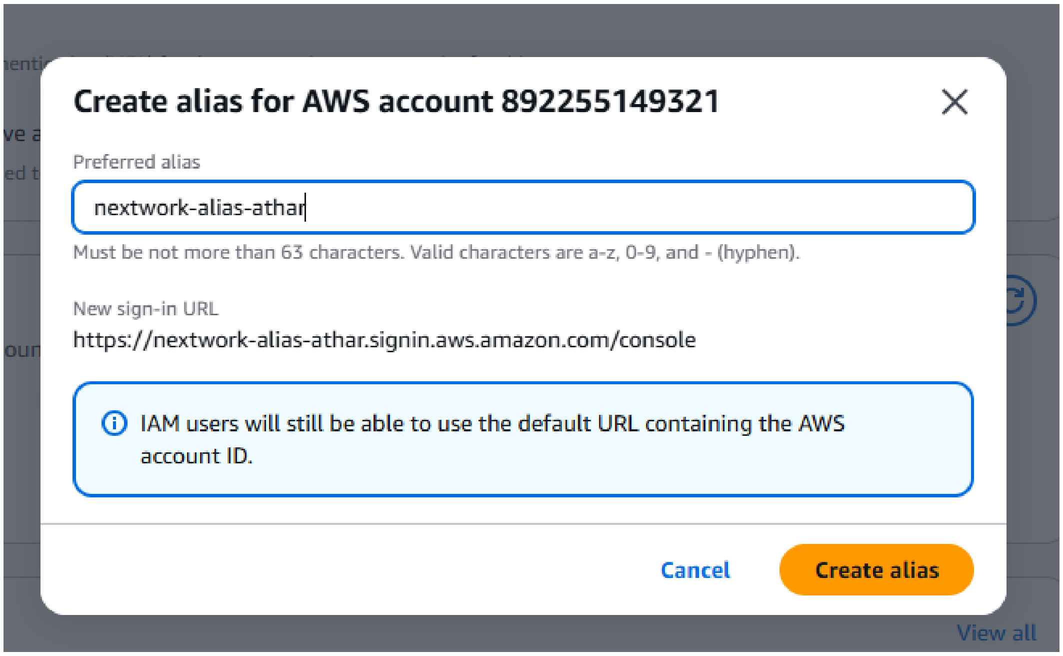

Figure 1: AWS Account Alias Creation (nextwork-alias-athar)

Step 2: Created EC2 instances for production and development environments with respective tags.

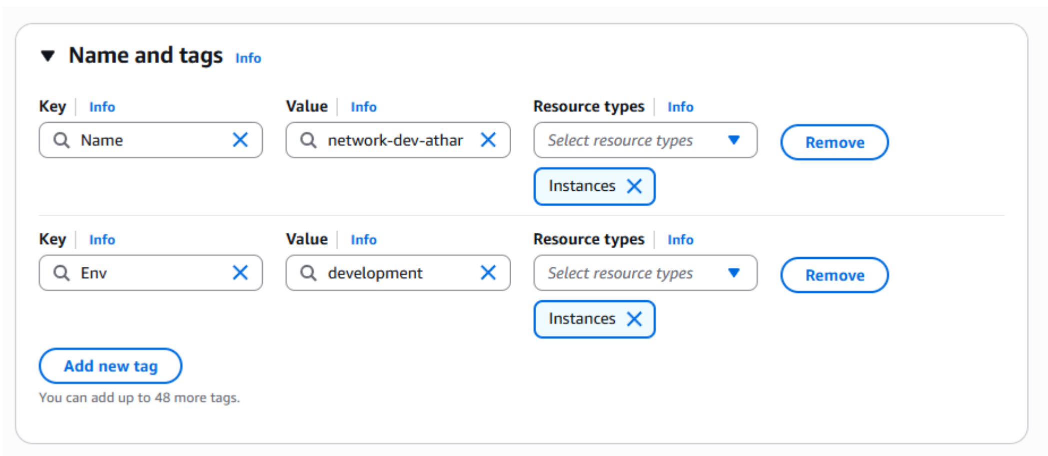

Figure 2: Development Instance Tagged with Env=development

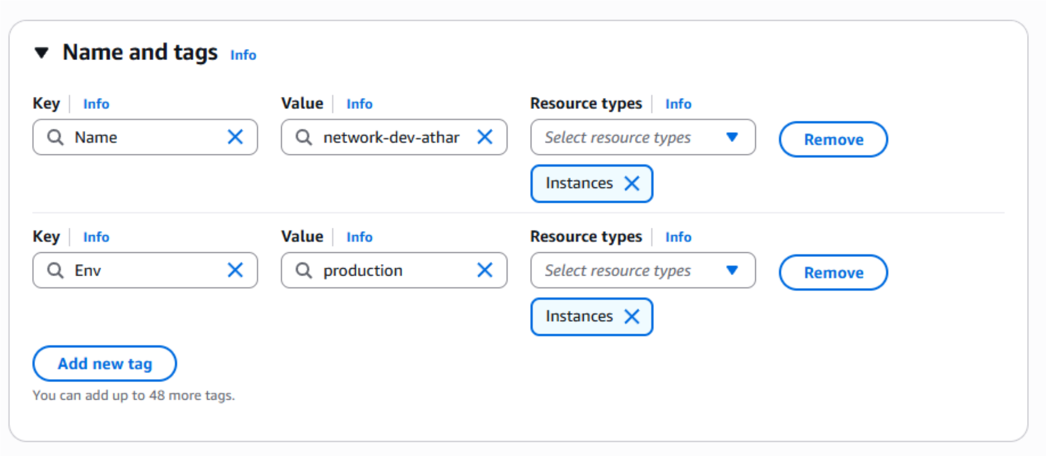

Figure 3: Production Instance Tagged with Env=production

Step 3: Created an IAM policy in JSON format to allow EC2 actions only on development instances.

Figure 4: IAM Policy JSON (NextWorkDevEnvironmentPolicy)

Step 4: Created IAM user and group, and assigned the policy.

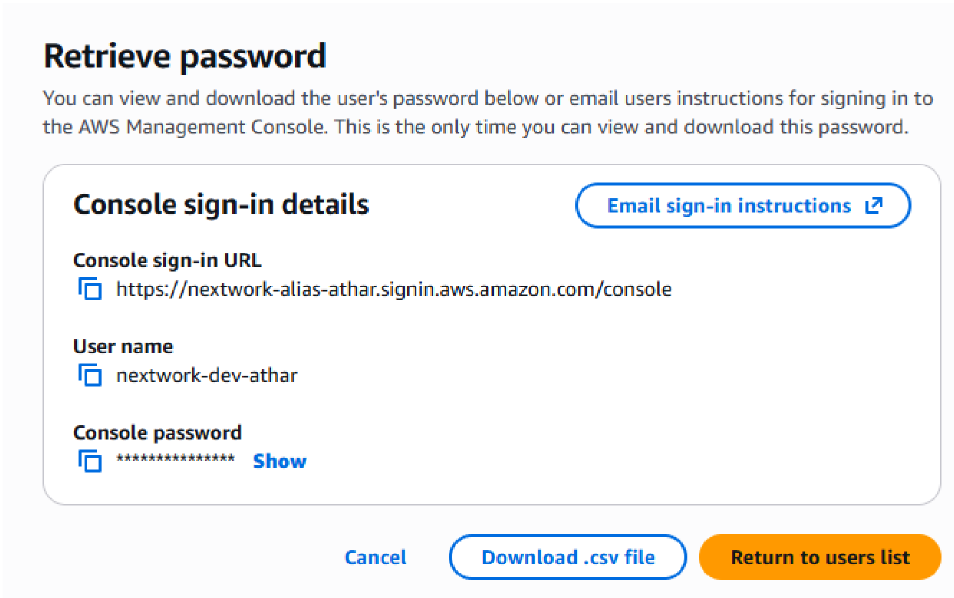

Figure 5: IAM User Creation for nextwork-dev-athar

5. Results

When testing user permissions, the IAM user could stop the development instance successfully but was denied access when trying to stop the production instance, demonstrating that the IAM policy worked correctly.

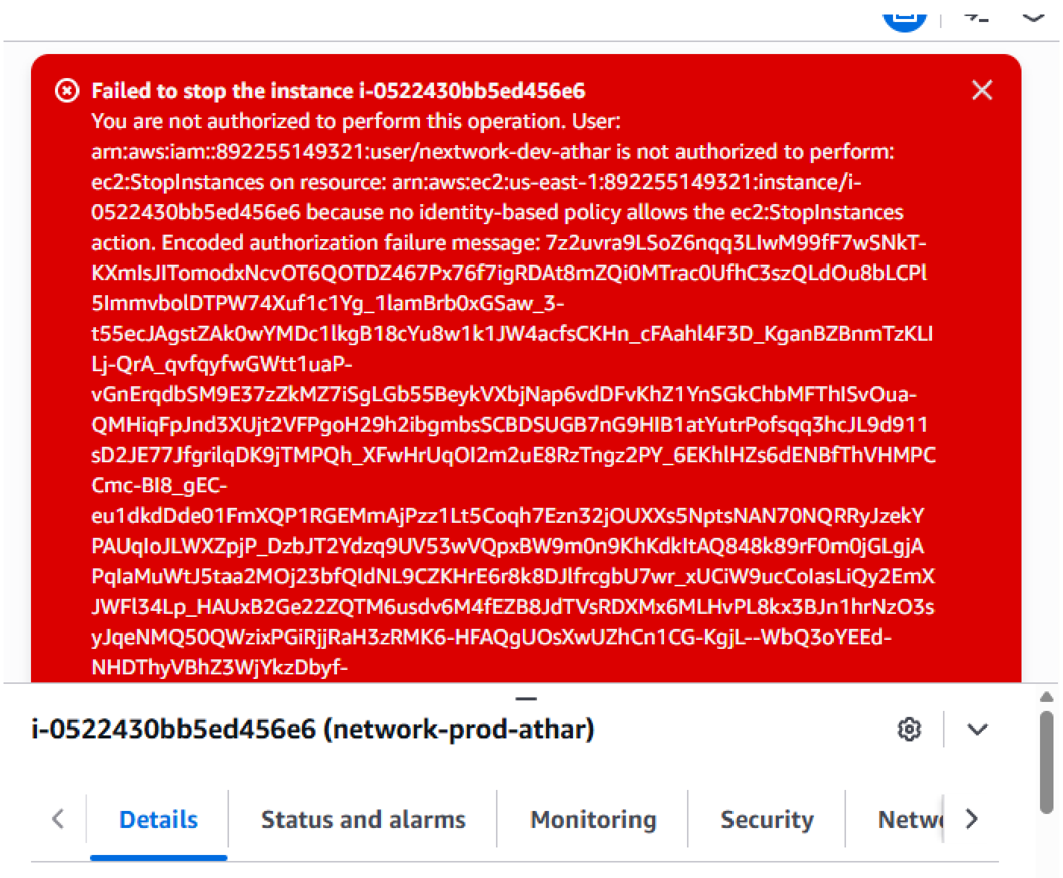

Figure 6: Access Denied – Attempt to Stop Production Instance

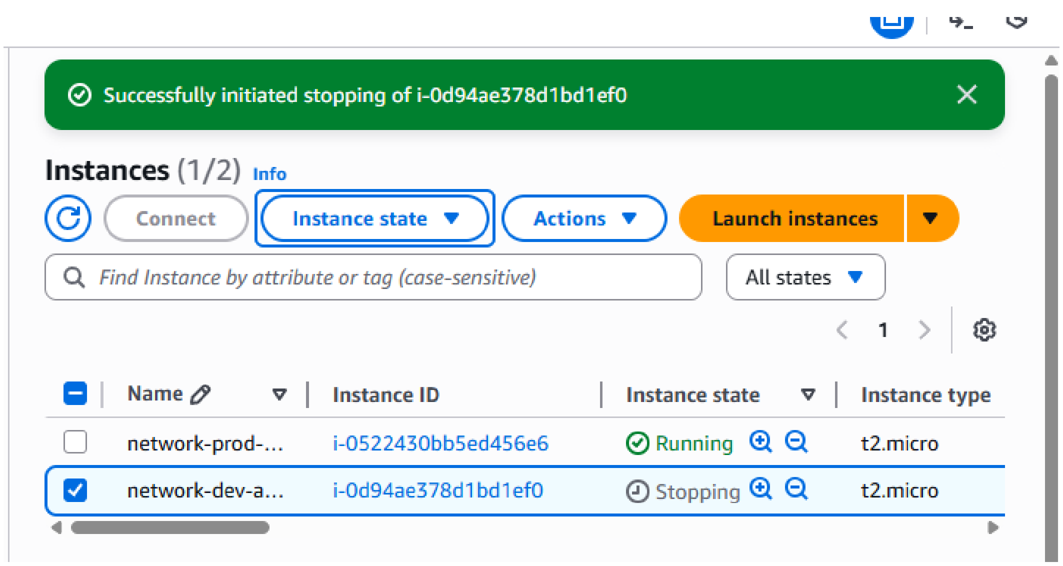

Figure 7: Successful Stop of Development Instance

6. Security Best Practices Applied

- Principle of Least Privilege – users only have access to what they need.
- Tag-based access control for environment-specific permissions.
- Group-based policy assignment for efficient management.
- Restricted tag manipulation to preserve resource integrity.

7. Conclusion

The AWS IAM and EC2 Access Control Project demonstrates effective implementation of role-based access control in a cloud environment. By using IAM policies, environment tags, and user groups, we successfully restricted access to production resources while allowing development flexibility. This project reinforces practical cloud security principles relevant to real-world DevOps and cybersecurity practices.

8. Future Enhancements

- Enable Multi-Factor Authentication (MFA) for all IAM users.
- Integrate AWS CloudTrail for access logging and auditing.
- Automate access testing using AWS CLI or Lambda functions.
- Introduce IAM roles for finer-grained permission control.